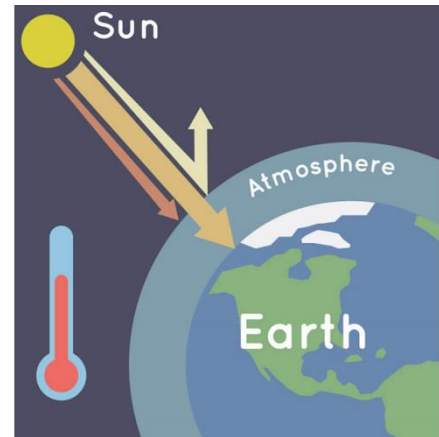


Global Climate Change

Greenhouse Effect

The earth's atmosphere constitutes several gases such as water vapor (H_2O), carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) that absorb and release heat, thus warming the atmosphere. These gases, are called greenhouse gases, and allow mostly visible light and a certain amount of infrared and ultraviolet (UV) radiation from the sun, to pass through the atmosphere. This is absorbed by the earth's surface, which transforms it into longer-wavelength infrared radiation (heat), which then rises into the lower atmosphere. Some of this heat escapes into space, while the rest are absorbed by these greenhouse gases and emitted into the lower atmosphere as even longer-wavelength infrared radiation. This natural warming effect of the troposphere is called the natural greenhouse effect, and is essential in maintaining the temperature of the earth's surface.



Earth's atmosphere traps some of the Sun's heat, preventing it from escaping back into space at night.
Credit: NASA/JPL-Caltech

Human activities such as burning fossil fuels, clearing forests and growing crops release carbon dioxide, methane and nitrogen oxide into the atmosphere in increasing amounts to such an extent that it has resulted in a significant increase in the average temperature of the earth. Global warming is defined as the human-enhanced warming of the atmosphere.

Global Climate Change

Climate change is the long-term shift in temperatures and weather patterns, that usually occur naturally over a period of time. But the last two hundred years have witnessed human activities to be the main cause of climate change, driven primarily by industrial activities. But a small change of even $1\text{-}2^\circ\text{C}$ can cause potentially dangerous shifts in the weather and climate patterns. These real, observable changes are what we designate as climate change impacts. The effects of climate change on the different aspects of the environment are discussed below.

1. More Frequent and Severe Weather

Warmer temperatures increase the frequency, intensity, and duration of heat waves, which can pose health risks, particularly for young children and the elderly. Warmer atmosphere can hold more moisture, which is eventually dumped back to the earth, resulting in extreme weather, including increasing number of droughts, intense storms, and floods. Drought conditions jeopardize access to clean drinking water, fuel wildfires, and result in dust storms, extreme heat events, and flash flooding. At the opposite end of the spectrum, heavier rains cause streams, rivers, and lakes to overflow, which damages life and property, contaminates drinking water, creates hazardous-material spills, promotes mold infestation and unhealthy air. A warmer, wetter world also promotes food-borne and waterborne illnesses and disease-carrying insects such as mosquitoes, fleas, and ticks.



2. Melting Ice Caps

When solar radiation hits snow and ice, approximately 90% of it is reflected back out to space. As global warming causes more snow and ice to melt each summer, the ocean and land that were underneath the ice are exposed at the Earth's surface. Because they are darker in color, the ocean and land absorb more incoming solar radiation, and then release the heat to the atmosphere. In this way, melting ice causes more warming and so more ice melts.



3. Melting Permafrost Releases Greenhouse Gases.

Global warming is causing soils in the polar regions that have been frozen for as much as 40,000 years to thaw. As they thaw, carbon trapped within the soils is released into the atmosphere as carbon dioxide and methane. These gases, released to the atmosphere, cause more warming, which then thaws even more of the frozen soil.

4. Higher Air Pollution

Rising temperatures also worsen air pollution by increasing ground level ozone, which is created when pollution from cars, factories, and other sources react to sunlight and heat. Ground-level ozone is the main component of smog, and the hotter things get, the more of it we have. Dirtier air is linked to higher hospital admission rates and higher death rates for asthmatics. It worsens the health of people suffering from cardiac or pulmonary disease. And warmer temperatures also significantly increase airborne pollen, which is bad news for those who suffer from hay fever and other allergies.



5. More Acidic Oceans

Oceans are becoming more acidic, due in large part to their absorption of some of our excess emissions. As this acidification accelerates, it poses a serious threat to underwater life, particularly creatures with calcium carbonate shells or skeletons, including molluscs, crabs, and corals. This can have a huge impact on shellfisheries and other industries that depend on the harvest of oysters, clams, and other shelled molluscs.

6. Rising Sea Levels

Global sea level has risen by about 8 inches since the year 1880, at a rate of 1-2 mm each year. This is the result of added water from melting land ice and the expansion of seawater as it warms. Polar regions are particularly vulnerable to a warming atmosphere. Average temperatures in the Arctic are rising twice as fast as they are elsewhere on earth. And it has been estimated that by the year 2100, our oceans will be 1-8 feet higher. This increase threatens coastal systems and low-lying areas, including entire island nations, including Some of the world's largest cities, including New York, Los Angeles, Miami, Mumbai,



Sydney, and Rio de Janeiro. For example, for a low-lying island nation like the Maldives in the Indian Ocean, even a small rise in sea levels could spell disaster for its people. About 80% of the 1,192 small islands making up this country lie less than 1 above sea level. Rising sea levels and higher storm surges during this century could flood most of these islands and their coral reefs. Next, let us talk about the increasing death rates due to climate change.

7. Higher Death Rates

Today's scientists point to climate change as "the biggest global health threat of the 21st century." It's a threat that impacts all of us—especially children, the elderly, low-income communities, and minorities—in a variety of direct and indirect ways. As temperatures spike, so does the incidence of illness, emergency room visits, and death. Climate change also has impacts on the wildlife species, both terrestrial and aquatic. Climate change has resulted in higher wildlife extinction rates.

8. Higher Wildlife Extinction Rates

As humans, we face a host of challenges, due to climate change, but we're certainly not the only ones catching the heat. As land and sea undergo rapid changes, the animals that inhabit them are doomed to disappear if they don't adapt quickly enough. Some will make it, and some won't. Many land, freshwater, and ocean species are shifting their geographic ranges to cooler climates or higher altitudes, in an attempt to escape climate warming. They're changing seasonal behaviours and traditional migration patterns as well. And yet, many face "increased extinction risk due to climate change." a 2015 study has revealed that vertebrate species—animals with backbones, like fish, birds, mammals, amphibians, and reptiles—are disappearing 114 times faster than they should be, a phenomenon that has been linked to climate change, pollution, and deforestation. Let us now understand the important factors that are responsible for these devastating climate change events. The first and foremost driver of climate change is the greenhouse effect, which is mostly driven by human activities.

Causes of Global Climate Change

1. Human Vs Natural Causes

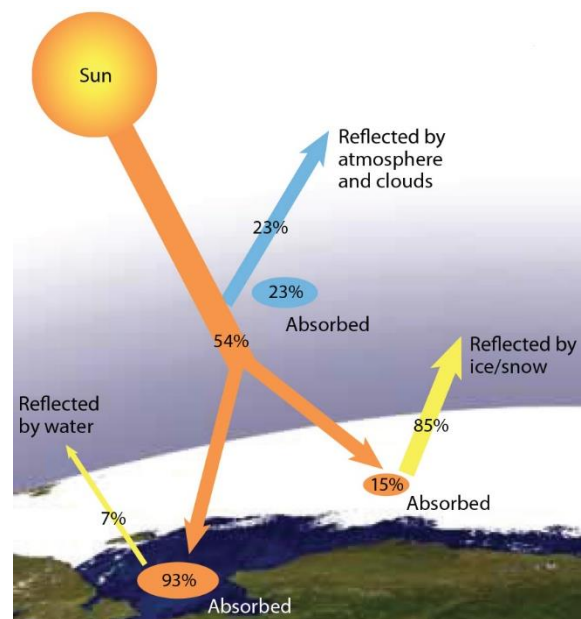
Scientists have pieced together a record of the earth's climate by analyzing a number of indirect measures of climate, such as ice cores, tree rings, glacier lengths, pollen remains, and ocean sediments, and by studying changes in the earth's orbit around the sun. This record shows that the climate varies naturally over a wide range of time scales. but this variability does not explain the warming that's been observed since the 1950s. it is extremely likely (> 95%) that human activities have been the dominant cause of this warming. CO₂ produced by human activities is the largest contributor to global warming. Methane is a more powerful greenhouse gas than CO₂, but has a shorter lifetime. Nitrous oxide is a long-lived greenhouse gas that accumulates in the atmosphere over decades to centuries. Currently, the CO₂ levels are at a record high of 414.8 ppm, a concentration that has not been seen on Earth for millions of years. Some of these activities include:

- Burning coal, oil and gas, producing carbon dioxide and nitrous oxide.
- Cutting down forests (deforestation). Trees help regulate the climate by absorbing CO₂ from the atmosphere. When cut down, that beneficial effect is lost and the carbon stored in the trees is in turn released into the atmosphere, adding to the greenhouse effect.

- Manufacturing and industry produce emissions, mostly from burning fossil fuels to produce energy for making things such as cement, iron, steel, electronics, plastics, clothes, and other goods. Mining and other industrial processes also release gases.
- Increasing livestock farming such as cows and sheep produce large amounts of methane when they digest their food.
- Fertilisers containing nitrogen produce nitrous oxide emissions.
- Fluorinated gases are emitted from equipment and products that use these gases. Such emissions have a very strong warming effect, up to 23,000 times greater than CO₂. Let us look at certain natural events that can affect the earth's surface temperature. The first factor is a change in the Reflectivity or Absorption of the Sun's Energy.

2. Reflectivity or Absorption of the Sun's Energy

Activities such as agriculture, road construction, and deforestation can change the reflectivity of the earth's surface, leading to local warming or cooling. This effect is observed in the form of heat islands, which are urban centres that are warmer than their surroundings, less populated areas. One reason for these effects is that urban centres house more buildings, pavements, and roofs that tend to reflect less sunlight than natural surfaces. Dark objects and surfaces, like the ocean, forests, and soil, tend to absorb more sunlight. Light-coloured objects and surfaces, like snow and clouds, tend to reflect more sunlight. About 70% of the sunlight that reaches the earth is absorbed. Natural changes in the earth's surface, like the melting of sea ice, have contributed to climate change in the past, often acting as feedbacks to other processes.



3. Changes in the Earth's Orbit and Rotation

Changes in the earth's orbit and its axis of rotation have had a big impact on climate in the past. For example, the amount of summer sunshine on the Northern Hemisphere, which is affected by changes in the planet's orbit, appears to be the primary cause of past cycles of ice ages, during which the earth has experienced long periods of cold temperatures (ice ages), as well as shorter interglacial periods (i.e., periods between ice ages) of relatively warmer temperatures.

4. Variations in Solar Activity

Variations in the sun's energy output can also affect the intensity of the light that reaches the earth's surface. Satellites have been measuring the amount of energy that the earth receives from the sun since 1978. And These measurements show no net increase in the sun's output, even as global surface temperatures have risen.

5. Volcanic Activity

Explosive volcano eruptions can throw particles (e.g., SO₂) into the upper atmosphere, where they can reflect enough sunlight back to space to cool the surface of the planet for several years. These particles are an example of cooling aerosols, which reflect the sunlight away from the

earth's surface. Volcanic particles from a single eruption do not produce long-term climate change because they remain in the atmosphere for a much shorter time than greenhouse gases.

6. Changes in Naturally Occurring Carbon Dioxide Concentrations

Over the last several hundred thousand years, carbon dioxide levels have varied in tandem with the glacial cycles. During warm interglacial periods, carbon dioxide levels were higher. During cool glacial periods, carbon dioxide levels were lower. These changing concentrations have acted as a positive climate feedback, amplifying the temperature changes caused by long-term shifts in the earth's orbit.